

Unit

3

Lesson

1

Beyond Imagination

Your Gesture-to-Sound Masterpiece!

Time Frame: [90 min]



Lesson Objectives:

By the end of this lesson students will be able to:

- Design a strong, protective box that holds all the electronic parts.
- Make their design user-friendly, easy to carry, and comfortable to use.
- Use the “Hole” tool in Tinkercad to create openings for buttons and electronic parts.
- Convert between centimeters, meters, and millimeters.
- Identify parallel, perpendicular, and intersecting lines in their design.
- Find the area of shapes by counting squares or using multiplication.
- Use area to plan and design parts that fit together properly.

Challenge questions of the lesson

1. How can we design a box that protects the electronics and is comfortable to use?
2. Where should we place holes for buttons and wires so everything fits just right?
3. How do we convert between centimeters and millimeters when measuring our design?
4. How does knowing the area of a shape help us design parts that fit together?

SEL Relation

- Responsible Decision-Making: Making thoughtful design choices to create a safe, effective, and user-friendly solution.
- Self-Management: Staying focused through a multi-step design and measurement process, including planning, revising, and building.
- Relationship Skills: Collaborating with teammates to share ideas, divide responsibilities, and solve design challenges together.
- Social Awareness: Designing with empathy by thinking about how the device will be used and how to make it accessible and comfortable for others.
- Self-Awareness: Recognizing personal

Engage



Teacher's Role:

- **Facilitator:** Guide the discussion about how the Arduino can replace the mystery box, focusing on the components and the role it plays in the design. Encourage students to think about the importance of keeping all parts secure and functional.
- **Questioner:** Ask open-ended questions like “What would happen if the Arduino and its components were left unprotected?” or “How can we design a case that is both protective and easy to carry?”
- **Connector:** Help students link their understanding of the Arduino to the task at hand—designing a protective case. Encourage them to consider how the Arduino interacts with the other components and how that influences the design.
- **Motivator:** Build excitement by showing how engineers use their knowledge to make electronic devices portable and durable.
- **Inclusion Advocate:** Reinforce the idea that the design should be easy to use for everyone and should protect the Arduino as well as any other connected components.

Expected From Students:

- **Active Listening:** Students will pay attention to the role of the Arduino and how it functions within the overall design.
- **Empathy:** Students will think about how to protect the Arduino while ensuring it remains easy for users to interact with.
- **Idea Generation:** Students will generate ideas to protect the Arduino and its components, including using a strong but lightweight material to create a portable case.
- **Participation in Discussion:** Students will engage with their peers and share ideas for the protective case design.
- **Creative Problem-Solving:** Students will sketch their designs for a protective case that will safeguard the Arduino while allowing the project to remain functional and portable.

Parts of the books included.



1. Set the Scene with Curiosity

- “Today, Ms. Sara and the students are holding the talking machine. But something’s not right... What do you think might go wrong if they leave the Arduino open and uncovered?”

2. Read the Story with Expression

- Use character voices and dramatic pauses—especially for Ms. Sara’s questions.
- Gesture while speaking (e.g., pretend to hold the Arduino and look worried).

3. Pause and Predict

- After Ms. Sara’s line: “But if we leave it like this, what could go wrong?” — pause and ask:
 - “What do you think will happen if the wires stay loose?”
 - “Has anything you made ever broken because it wasn’t protected?”

4. Encourage Connection to Real-World Design

- “Engineers often have to protect the smart brains of devices. Today, you’ll be thinking like engineers to design a protective case for the Arduino—one that’s safe and easy to carry!”

Possible Strategies for Implementation:

1. Use Physical Props or Real Objects

- Hold up an actual Arduino board (or a printed image of one) to mimic Ms. Sara’s actions in the story.
- Ask students: “What could happen if this dropped on the floor?”

Helps students visualize and relate the problem to real-world concerns.

2. Turn the Story into a Mini Skit

- Assign roles to students (Ms. Sara, Adam, Lila, Omar) and act out the dialogue.
- Let students read aloud with expression and simple gestures.

Promotes engagement, listening, and ownership of the problem.

3. Think-Pair-Share Before Drawing

- After the question “What could go wrong?” have students:
 - Think quietly
 - Discuss with a partner
 - Share their ideas with the class

Builds speaking and listening skills and supports all learners in formulating responses.

4. Make the Problem Visual with Quick Sketches

- On the board, draw two versions:
 - A messy circuit with wires sticking out
 - A neat device safely inside a case
 - Ask: “Which one would you want to carry in your backpack?”

Great for visual learners and concrete thinking.

5. Connect to Prior Knowledge

- Ask: “Where else have you seen devices in a case or cover?” (e.g., phones, tablets, glasses)
- Guide them to realize why cases matter—and that engineers must think about this, too.

Activates everyday experiences to deepen understanding.

Let's Think!



Objective:

Students will brainstorm and visually represent a design for a protective and accessible case that holds electronic components, considering durability and ease of use for diverse users.

Possible Answers:

- “The case should be made of hard plastic so it won’t break.”
- “We can add foam or padding inside so the wires don’t move around.”
- “Let’s use a handle or strap so it’s easier to carry.”



A photo of what might students draw.

- “We can make it colorful or add symbols so people know what each button does.”
- “Let’s design it like a lunchbox—it opens and closes easily.”

Possible Strategies for Implementation:

1. Use Real-World Examples for Inspiration

- Show students everyday cases (e.g., phone cases, pencil boxes, eyeglass cases).
- Ask: “What makes this case strong? Easy to use?”

2. Encourage Pair Sketching

- Allow students to sketch in pairs or small groups and explain their ideas aloud.
- Promotes peer discussion and idea exchange.

3. Model a Sample Design

- Quickly draw a simple case example on the board to spark ideas.
- Label key features (handle, padding, labels).

4. Use Question Prompts to Spark Design Thinking

- Ask guiding questions like:
 - “What if it falls off a table?”
 - “How can we carry it with one hand?”
 - “What if someone with vision problems needs to use it?”

5. Support All Learners with Design Vocabulary Cards

- Provide visual cards with words like “handle,” “button,” “soft grip,” “label,” etc.
- Helps ELLs and visual learners express ideas more confidently.

Explorer’s Toolkit



Teacher’s Role

- Introduce the challenge by explaining that accurate measurements are key to successful 3D modeling.
- Demonstrate how to measure an object’s width, length, and height using a ruler.
- Model converting centimeters to millimeters on the board.
- Guide students in recording their data and using the area formula (length × width).
- Support struggling students by helping with unit conversions or formula understanding.
- Encourage critical thinking by asking why accurate measurement is important in design.

Expected From Students

- Measure the dimensions of their prototype or a sample talking device.
- Record width, length, and height in centimeters.
- Convert the measurements into millimeters.
- Calculate the area (bottom space) using the formula.
- Reflect on why understanding the exact space is essential for case design.
- Begin planning how the case might look in Tinkercad.

Parts of the books included



Let's Think! 

How can we design a case that protects all the parts inside, even if it gets dropped — and what can we add to make it easier to use by everyone?

"Sketch your ideas"

Explorer's Toolkit 

DIY Lab 

Today, you're going to act like real designers! Your mission is to create a 3D case for your talking device using Tinkercad — but first, you need to measure the real thing.



Measurement	Size (in cm)
Width	
Length	
Height	







Don't forget — Tinkercad measures everything in millimeters (mm)!
Need a clue?

- 1 meter = 100 centimeters
- 1 centimeter = 10 millimeters

So... how can you turn your measurements into millimeters to make sure your design fits perfectly?

Now that you have measured the length and width of your device, use those numbers to calculate the area of its bottom.

Use this formula: Bottom area = length × width

Why do you think it's important to know how much space your device takes up before designing a case for it in Tinkercad?

Screen Lessons 

Watch the video learn more about area.

<https://www.youtube.com/watch?v=bH601rApG4>





Figure 30: A character exploring the concept of area.







DIY Lab

Objective:

Students will measure real-world dimensions of their talking device and calculate area to design a custom-fit 3D case in Tinkercad using metric units (mm and cm).

Materials Needed:

- Ruler (marked in cm)
- Pencil
- Measurement worksheet in the book
- Sample of the talking device
- Tinkercad access (for later 3D modeling)

Activity Instructions:

- **Introduce the Challenge.**
 - Tell students they are going to act like real product designers: “We’re designing a case for our talking device—but first, we need to know its exact size.”
- **Model the Measurement Process.**
 - Hold up a ruler and show how to measure length, width, and height of a sample object.
 - Emphasize aligning the ruler correctly and reading the cm scale.
 - Demonstrate how to convert centimeters to millimeters (e.g., 1 cm = 10 mm).
- **Guide Students in Recording Data**
 - Walk students through filling out the table in their worksheet.
 - Pause and check for understanding after each measurement step.
- **Support Area Calculation**
 - On the board or projector, model how to calculate area using $\text{Area} = \text{length} \times \text{width}$.
 - Encourage students to use their measured values to find the bottom space their case needs.
- **Lead a Reflection**
 - Ask students: “Why do you think measuring first is important when you design in Tinkercad?”
 - Facilitate a short class discussion and encourage them to write a sentence about it on their worksheet.

Possible Answers:

- **Sample Measurement:**
 - Width = 6 cm → 60 mm
 - Length = 10 cm → 100 mm
 - Height = 3 cm → 30 mm
- **Area = $10 \times 6 = 60 \text{ cm}^2$**
- **Reflection:**

“If I don’t measure right, the device might not fit. The case could be too tight or too loose. Knowing the exact size helps me build it correctly in Tinkercad.”

Possible Strategies for Implementation:

1. Use a Measurement Demo Station

- Set up a demo table with a ruler, Arduino, and sample objects.
- Let students take turns practicing measuring with support before doing it on their own.
- Reinforces proper tool use and reduces measurement errors.

2. Peer Measuring Teams

- Pair students to measure together—one holds the device, one uses the ruler.
- They take turns recording measurements.
- Promotes collaboration and improves accuracy.

3. Visual Measurement Posters

- Display visual aids showing how to align a ruler, read centimeters, and convert to millimeters.
- Include formula visuals like: $\text{Area} = \text{length} \times \text{width}$.
- Supports visual learners and builds math confidence.

4. Tinkercad Integration Preview

- Open Tinkercad and show how measurements are input digitally.
- Help students understand why precise measurements matter before they design.
- Builds a real-world connection between math and design.

5. Think-Pair-Share Reflection

- After completing measurements, ask:
 - “What would happen if we didn’t measure correctly?”
 - “Why does space matter in designing a case?”
- Let students discuss in pairs and share with the class to deepen understanding.

6. Optional: Use Grid Paper for Practice

- Provide grid paper to visualize the base area ($\text{length} \times \text{width}$) before entering it in Tinkercad.
- Supports spatial reasoning and pre-design thinking.

Explain



Teacher's Role:

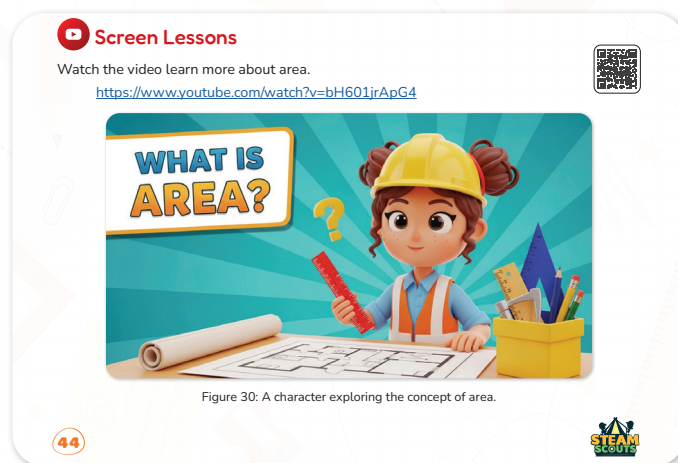
- Facilitator: Introduce the concept of area and connect it to real-life applications like building or designing objects.
- Guide: Help students focus on key ideas while watching the video.
- Connector: Relate what they learn about area to the hands-on activity of designing a case in Tinkercad.
- Encourager: Support students in asking questions and applying math confidently.

Expected From Students:

- Watch the video attentively and identify how area is measured.
- Connect the video's concepts to their own measurement task.

- Ask questions or share something new they learned.
- Use the formula (length × width) in their upcoming design activity.

Parts of the books included.



Screen Lessons

Objective:

Students will understand the concept of area and how to calculate it using length × width. This supports their ability to measure and plan the space needed for designing a case for their device.

Video Link: <https://www.youtube.com/watch?v=bH601jrApG4>



Possible Strategies for Implementation:

1. Pre-Watching: Set a Purpose

- Strategy: Math in Design Connection
 - Say: “We’re about to design something real—a case for your device. But to do that, we need to know how much space it takes. That’s called area. Let’s learn what that means.”
 - Ask: “Have you ever measured a table or a book before? What do you think ‘area’ means?”

2. During Watching: Active Engagement

- Prepare a simple sheet with questions to answer while watching:
 - What does area mean?
 - What is the formula for area?
 - What is one thing from the video you didn’t know before?
- Pause the video at key points to highlight examples or ask clarifying questions.

3. Post-Watching: Discussion and Reflection

- Strategy 1: Real-World Examples
 - Ask: “Where do we use area in real life?”

- (Possible answers: rug size, room space, phone screen, etc.)
- Strategy 2: Peer Practice Challenge
 - Give students two sample rectangles and ask them to calculate the area.
 - Discuss: “Why is knowing the area important when we design things in Tinkercad?”
- Strategy 3: Area Sketch Walk
 - Have students draw a quick sketch of their device base on grid paper and label the length, width, and area.

Elaborate



Teacher's Role

- Math Guide: Introduce the area activity by reviewing how to count square units. Demonstrate how to visually estimate and verify area by counting colored blocks.
- Facilitator: Support students in independently or collaboratively solving area problems. Encourage students to double-check their answers.
- Design Coach: Guide students through the Tinkercad case design by showing how geometry helps in creating functional 3D objects.
- Geometry Connector: Discuss real-world examples of parallel, perpendicular, and intersecting lines in product design. Prompt students to identify these features in their own models.
- Encourager: Celebrate creative ideas and neat geometry use, reinforcing that there are many right ways to design strong and useful cases.

Expected From Students

- Apply Area Concepts: Count square units accurately to find the area of different shapes. Circle the correct answers confidently.
- Geometry Reflection: Identify where they used lines and shapes in their Tinkercad case and explain their design choices using geometry vocabulary.
- Design with Purpose: Use Tinkercad tools to create a protective case for their Arduino-based talking device, ensuring proper fit and accessibility.
- Math Integration: Apply math knowledge (area and geometry) in a practical design task to make learning meaningful and connected.
- Explain Thinking: Use the worksheet to reflect on where geometry appears in their case and why those choices help the function or strength of the design.

Parts of the books included.



Apprentice

Objective:

Students will practice identifying and calculating the area of shapes by counting square units. They will reinforce their understanding of how area is measured and represented using visual models.

Answers:

- Red shape (2 squares): ☒ 2
- Blue shape (6 squares): ☒ 6
- Green shape (4 squares): ☒ 4
- Yellow shape (3 squares): ☒ 3

Activity Instructions for Teachers:

1. Introduce the Concept:

- Remind students that area is the amount of space a shape covers, and we measure it by counting the square units inside the shape.

2. Model a Sample:

- Demonstrate how to count squares for one example on the board (e.g., the red shape).

3. Guide the Activity:

- Distribute the worksheet.
- Ask students to count the squares in each shape and circle the correct answer from the options provided.

4. Check Understanding:

- Circulate the room to observe counting strategies.
- Ask students to explain how they got their answers.

Possible Strategies for Implementation:

1. Use a Visual Anchor:

- Display a large shape on the board and model how to count each square unit, tapping or highlighting each one.

2. Partner Check:

- Have students work in pairs to solve, then cross-check each other's answers.

3. Number Talk:

- After the activity, ask questions like:
 - "Which shape had the most squares?"
 - "How did you count — by rows, columns, or individually?"

4. Link to Real-World Contexts:

- Ask students, "Why might it be important to know the area when designing something like a robot case or a poster?"

5. Early Finisher Challenge:

- Let fast finishers create their own shape with square units and swap with a friend to calculate the area.



Builder

Objective: Students will apply their knowledge of 3D design and geometry to create a protective and functional case for their talking device using Tinkercad. They will identify and label geometric features such as parallel, perpendicular, and intersecting lines within their design.

Geometry Chart – Sample Answers

Geometry Concept	Where did you use this in your case design?
Parallel Lines	The top and bottom edges of the case / sides of button slots
Perpendicular Lines	Where the front and side walls meet at a corner / base and side edges
Intersecting Lines	Where the holes for the speaker or button cutouts meet the main surface

Activity Instructions

1. Introduce the Design Task:

- Explain that the students will use Tinkercad to design a case that protects their Arduino talking device and makes it easy to carry and use.

2. Discuss Key Features:

- Show the example case.

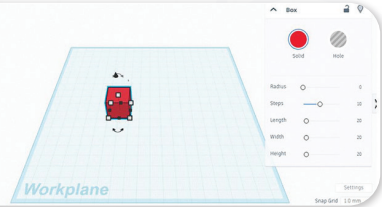
- Highlight important design considerations:
 - Rounded edges or corners for safety
 - Openings for speaker/sound
 - Button placements
 - Solid base for durability

3. Students Begin Designing:

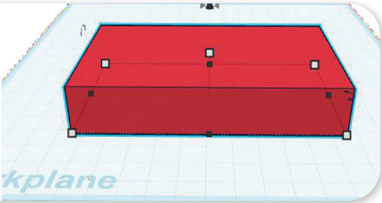
- Guide students to open Tinkercad and create a new design.
- Encourage them to apply their measurements from the Explore phase to size the case correctly.

Step by step guide for the design:

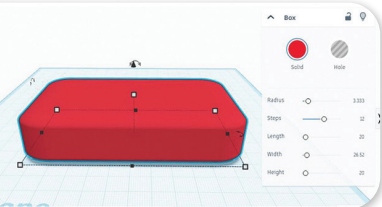
1. Grap a cube from the basic shapes



2. Change the width and Hight from the black squares



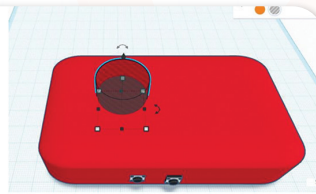
3. To make the bevel change the radios and the steps from the steps



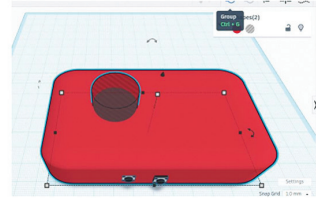
5. Rotate the push buttons 90 degrees in z axis from the two headed arrow then insert them into the box



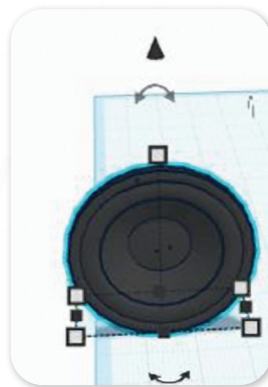
6. Garp the hollow cylinder and insert it into the cube



7. Select the box and the cylinder then group them from the group icon or using ctrl+g to create a hole in the box



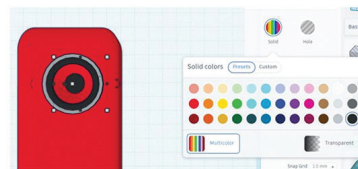
8. From the link copy the speaker and past it into your design then scale it down



10. Place the cylinder into the hole you created



11. From the properties menu choose multicolor to bring the speaker to it's original colors



4. Introduce Geometry Vocabulary Task:

- Once the design is created, students complete the geometry chart by identifying:
 - Parallel lines (e.g., top and bottom edges)
 - Perpendicular lines (e.g., corners)
 - Intersecting lines (e.g., cross-sections or interior design features)

Possible Strategies for Implementation:

1. Model the Process Live:

- Open Tinkercad on a projector and walk students through how to add shapes, rotate, resize, and group objects to form a case.

2. Geometry Hunt Challenge:

- Turn the vocabulary task into a scavenger hunt: “Can you find one example of each geometry term in your design?”

3. Use Design Checklists:

- Provide a printed or digital checklist for students to ensure their case has all required elements before submitting.

4. Peer Review:

- Pair students to review each other’s Tinkercad cases and spot the geometry features.

5. Anchor Posters:

- Post visuals of parallel, perpendicular, and intersecting lines in the classroom for easy reference.

6. Prompt Reflection:

- Ask: “How did geometry help you make a better case? What part of your case would be hardest to build in real life?”

Evaluate



Teacher’s Role

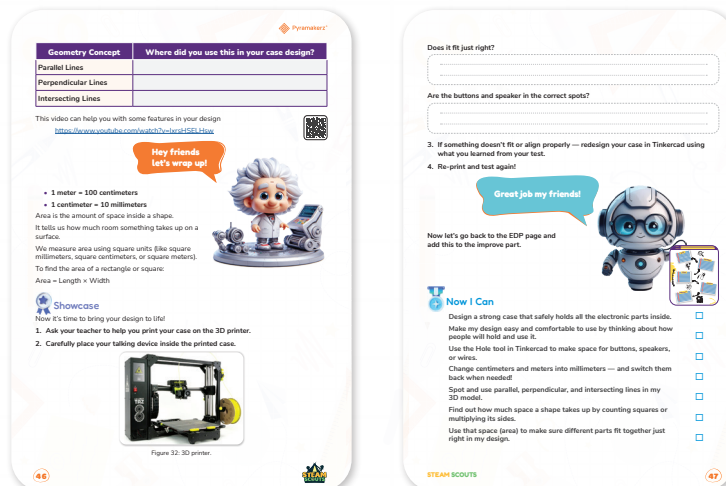
- Facilitator of Real-World Testing
- Guide students through the process of testing their 3D-printed cases, encouraging reflection and iteration.
- Technical Support
- Assist with placing devices into cases, managing fragile components, and helping students observe alignment.
- Questioner and Reflective Guide
- Pose guiding questions like “Does it fit just right?” or “What would you change?” to promote critical thinking and design reasoning.
- Encourager of Improvement
- Reassure students that mistakes are part of the process. Encourage redesign when needed and celebrate persistence and learning.
- Showcase Organizer
- Create a supportive environment for testing and sharing—set up a test table or walk-

through area for small groups to present.

Expected from Students

- Careful Observation
- Students will place their Arduino or device into the 3D-printed case and check for proper fit and alignment.
- Critical Evaluation
- They will identify if button/speaker holes align correctly and whether the case fits snugly or needs adjustments.
- Documentation
- Students will record observations directly on their worksheet, answering if the design worked and what needs improvement.
- Collaboration & Feedback
- Students may share designs with classmates, compare results, and suggest or discuss changes for improvement.
- Iteration & Growth Mindset
- If something doesn't work, students are expected to reflect, redesign, and try again—mirroring how real engineer's work.

Parts of the books included



Objective: Students will test and evaluate their 3D-printed case design by checking its fit, alignment, and function, and then iterate on the design using feedback from real-world testing.

Materials Needed:

- Assembled talking device (with Arduino, buttons, speaker, etc.)
- Access to Tinkercad
- 3D printer

Activity Instructions:

1. Prepare 3D Printed Cases

- Ensure all student designs have been exported from Tinkercad and sliced for 3D printing.
- Use UltiMaker Cura to help you do this step.

Download link: <https://ultimaker.com/software/ultimaker-cura/>

Cura settings: https://drive.google.com/file/d/12_2KEokjiqC0-wWx-xItlrCs96nV5Uj4/view?usp=drive_link



- Print student cases before the class or during group rotations if time permits.

2. Introduce the Goal

- Remind students that engineers test their prototypes to make sure everything works before final production.
- Explain that today's goal is to test how well their designed case protects and fits the device.

3. Organize a "Test Station"

- Set up a table with the 3D-printed cases and pre-built Arduino-based talking devices (or dummy models).
- Call small groups to test one at a time to ensure careful handling.

4. Facilitate the Fit Check

- Ask guiding questions as students place the device into their case:
 - "Does the Arduino fit without moving?"
 - "Are the speaker and button holes in the right place?"
 - "Is it easy to carry or hold?"

5. Support Redesign Conversations

- If something doesn't work, prompt students to reflect:
 - "What part needs fixing?"
 - "Should you move or resize something in Tinkercad?"
- Encourage them to sketch a revised idea or reopen Tinkercad if time allows.

6. Document Observations

- Ask students to record their answers on the printed worksheet (fit and alignment).
- Have students share with a partner or do a "gallery walk" to compare designs.

7. Celebrate Iteration

- Remind students it's okay if the first version isn't perfect—real engineers test and improve!
- Highlight examples of improvement and persistence.

Possible Strategies for Implementation:

1. Test-Fit Stations:

- Set up a station where each student gets a turn with the 3D-printed case to try their fit and alignment. Use a checklist to guide testing.

2. Peer Feedback:

- Let students test each other's cases and give feedback: "Does this speaker hole match your design?" "Do the buttons press easily?"

3. Guided Redesign Walkthrough:

- Walk through a live example of a design that didn't fit — model how to make adjustments in Tinkercad (e.g., moving button hole 1 cm left).

4. Mini Design Journal Prompt:

- Ask students to write 1–2 sentences on what worked and what they would fix. This supports reflection and iterative thinking.

5. Celebrate Prototypes:

- Create a "Device Gallery Walk" where students showcase working cases and explain how they made improvements based on testing.

Final prototype:



Final prototype video: <https://drive.google.com/drive/folders/1VPeZflpX6metwdl7Wm-q1pifLUrfmRg5>



Rubric for Evaluating

It's time to move to the improve part and add this.

"In case of project fit they will write that. In case it didn't fit they will write the response below."



Criteria	Excellent (4)	Good (3)	Needs Improvement (2)	Beginning (1)
Fit and Alignment	All parts fit perfectly, no gaps or misalignment	Mostly fits with minor adjustments needed	Some components don't fit properly	Device does not fit at all
Redesign and Iteration	Made clear, thoughtful improvements after testing	Made some improvements based on testing	Attempted improvements, but not effective	Did not attempt to redesign
Functionality of Case	Device operates fully inside the case (buttons/sound)	Mostly functional, slight issues	Buttons or sound do not work properly	Case prevents device function
Presentation and Reflection	Clearly explains design changes and what was learned	Explains basic changes and outcome	Struggles to explain design choices	No explanation or unclear reasoning



Now I Can

Objective:

Students will reflect on and assess their learning by identifying key skills and concepts they've practiced—specifically related to 3D design, spatial measurement, unit conversion, geometry, and functional prototyping using Tinkercad.

Activity Instructions for Teachers:

1. Introduce the Checklist:

- Say: "Let's look back and see all the amazing things you've learned to do as real

engineers!”

- Project or display the checklist for all students.

2. Review Each Statement Together:

- Read each sentence aloud and briefly explain or model it if needed.
- Pause after each one to allow students to recall when they used that skill.

3. Student Reflection:

- Ask students to check the boxes for each skill they feel confident in.
- Encourage them to put a star next to skills they want to learn more about or improve next time.

4. Optional Wrap-Up Discussion:

- Facilitate a class conversation:
 - “Which skill was your favorite?”
 - “What was the trickiest part?”
 - “How did using math help you in your design?”

Relation to Standards

1. NGSS – Next Generation Science Standards

3-5-ETS1-1:

Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials.

(Students design a custom protective case that fits measured specifications and meets usability requirements.)

3-5-ETS1-2:

Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints.

(Students explore different design features such as shape, button placement, and durability in Tinkercad.)

2. CCSS – Common Core State Standards (Mathematics)

3.MD.C.5:

Recognize area as an attribute of plane figures and understand concepts of area measurement.

(Students calculate surface area for design accuracy.)

3.MD.C.7:

Relate area to the operations of multiplication and addition.

(Students use multiplication to find the area of their device’s base.)

3.G.A.1:

Understand shapes in different categories and recognize shared attributes (e.g.,

quadrilaterals, parallel and perpendicular lines).

(Students apply knowledge of geometry in creating functional, structured designs.)

3. ISTE – International Society for Technology in Education

1.4 Innovative Designer:

Students use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions.

(Designing a digital 3D case using Tinkercad that addresses both form and function.)

1.5 Computational Thinker:

Students develop strategies for understanding and solving problems using technology.

(Students apply logic, measurement, and geometry in a digital modeling environment.)